1. Import Libraries

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

1. Load Dataset

```
data = pd.read_csv("hotel_bookings.csv")

#first 5 rows Show
data.head()
```

```
   index         hotel  is_canceled  lead_time  arrival_date_year  \
0      0  Resort Hotel            0        342               2015
1      1  Resort Hotel            0        737               2015
2      2  Resort Hotel            0          7               2015
3      3  Resort Hotel            0         13               2015
4      4  Resort Hotel            0         14               2015

  arrival_date_month  arrival_date_week_number
arrival_date_day_of_month  \
0               July                        27
1
1               July                        27
1
2               July                        27
1
3               July                        27
1
4               July                        27
1

   stays_in_weekend_nights  stays_in_week_nights  ...  deposit_type
agent  \
0                        0                     0  ...    No Deposit
NaN
1                        0                     0  ...    No Deposit
NaN
2                        0                     1  ...    No Deposit
NaN
3                        0                     1  ...    No Deposit
304.0
4                        0                     2  ...    No Deposit
240.0

   company days_in_waiting_list customer_type   adr  \
0      NaN                    0     Transient   0.0
1      NaN                    0     Transient   0.0
2      NaN                    0     Transient  75.0
```

```
3      NaN                     0     Transient  75.0
4      NaN                     0     Transient  98.0

  required_car_parking_spaces  total_of_special_requests
reservation_status  \
0                            0                          0
Check-Out
1                            0                          0
Check-Out
2                            0                          0
Check-Out
3                            0                          0
Check-Out
4                            0                          1
Check-Out

   reservation_status_date
0                 01-07-15
1                 01-07-15
2                 02-07-15
3                 02-07-15
4                 03-07-15

[5 rows x 33 columns]
```

1. Basic Understanding

```
# rows aur columns
data.shape
```

```
(119390, 33)
```

```
# columns check
data.columns
```

```
Index(['index', 'hotel', 'is_canceled', 'lead_time',
'arrival_date_year',
       'arrival_date_month', 'arrival_date_week_number',
       'arrival_date_day_of_month', 'stays_in_weekend_nights',
       'stays_in_week_nights', 'adults', 'children', 'babies', 'meal',
       'country', 'market_segment', 'distribution_channel',
       'is_repeated_guest', 'previous_cancellations',
       'previous_bookings_not_canceled', 'reserved_room_type',
       'assigned_room_type', 'booking_changes', 'deposit_type',
'agent',
       'company', 'days_in_waiting_list', 'customer_type', 'adr',
       'required_car_parking_spaces', 'total_of_special_requests',
       'reservation_status', 'reservation_status_date'],
      dtype='str')
```

```
# data types
data.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 119390 entries, 0 to 119389
Data columns (total 33 columns):
 #   Column                          Non-Null Count   Dtype
---  ------                          --------------   -----
 0   index                           119390 non-null  int64
 1   hotel                           119390 non-null  str
 2   is_canceled                     119390 non-null  int64
 3   lead_time                       119390 non-null  int64
 4   arrival_date_year               119390 non-null  int64
 5   arrival_date_month              119390 non-null  str
 6   arrival_date_week_number        119390 non-null  int64
 7   arrival_date_day_of_month       119390 non-null  int64
 8   stays_in_weekend_nights         119390 non-null  int64
 9   stays_in_week_nights            119390 non-null  int64
 10  adults                          119390 non-null  int64
 11  children                        119386 non-null  float64
 12  babies                          119390 non-null  int64
 13  meal                            119390 non-null  str
 14  country                         118902 non-null  str
 15  market_segment                  119390 non-null  str
 16  distribution_channel            119390 non-null  str
 17  is_repeated_guest               119390 non-null  int64
 18  previous_cancellations          119390 non-null  int64
 19  previous_bookings_not_canceled  119390 non-null  int64
 20  reserved_room_type              119390 non-null  str
 21  assigned_room_type              119390 non-null  str
 22  booking_changes                 119390 non-null  int64
 23  deposit_type                    119390 non-null  str
 24  agent                           103050 non-null  float64
 25  company                         6797 non-null    float64
 26  days_in_waiting_list            119390 non-null  int64
 27  customer_type                   119390 non-null  str
 28  adr                             119390 non-null  float64
 29  required_car_parking_spaces     119390 non-null  int64
 30  total_of_special_requests       119390 non-null  int64
 31  reservation_status              119390 non-null  str
 32  reservation_status_date         119390 non-null  str
dtypes: float64(4), int64(17), str(12)
memory usage: 30.1 MB
```

```
# statistical summary
data.describe()
```

```
                index    is_canceled      lead_time  arrival_date_year
\
count  119390.000000  119390.000000  119390.000000      119390.000000
```

```
mean    59694.500000       0.370416     104.011416        2016.156554

std     34465.068657       0.482918     106.863097           0.707476

min         0.000000       0.000000       0.000000        2015.000000

25%     29847.250000       0.000000      18.000000        2016.000000

50%     59694.500000       0.000000      69.000000        2016.000000

75%     89541.750000       1.000000     160.000000        2017.000000

max    119389.000000       1.000000     737.000000        2017.000000


       arrival_date_week_number  arrival_date_day_of_month  \
count             119390.000000              119390.000000
mean                 27.165173                  15.798241
std                  13.605138                   8.780829
min                   1.000000                   1.000000
25%                  16.000000                   8.000000
50%                  28.000000                  16.000000
75%                  38.000000                  23.000000
max                  53.000000                  31.000000

       stays_in_weekend_nights  stays_in_week_nights         adults  \
count            119390.000000         119390.000000  119390.000000
mean                 0.927599              2.500302       1.856403
std                  0.998613              1.908286       0.579261
min                  0.000000              0.000000       0.000000
25%                  0.000000              1.000000       2.000000
50%                  1.000000              2.000000       2.000000
75%                  2.000000              3.000000       2.000000
max                 19.000000             50.000000      55.000000

            children  ...  is_repeated_guest
previous_cancellations  \
count  119386.000000  ...      119390.000000           119390.000000

mean        0.103890  ...           0.031912                0.087118

std         0.398561  ...           0.175767                0.844336

min         0.000000  ...           0.000000                0.000000

25%         0.000000  ...           0.000000                0.000000

50%         0.000000  ...           0.000000                0.000000

75%         0.000000  ...           0.000000                0.000000
```

```
max        10.000000  ...           1.000000                26.000000


       previous_bookings_not_canceled  booking_changes          agent
\
count                   119390.000000    119390.000000  103050.000000

mean                         0.137097         0.221124      86.693382

std                          1.497437         0.652306     110.774548

min                          0.000000         0.000000       1.000000

25%                          0.000000         0.000000       9.000000

50%                          0.000000         0.000000      14.000000

75%                          0.000000         0.000000     229.000000

max                         72.000000        21.000000     535.000000


           company  days_in_waiting_list            adr  \
count  6797.000000         119390.000000  119390.000000
mean    189.266735              2.321149     101.831122
std     131.655015             17.594721      50.535790
min       6.000000              0.000000      -6.380000
25%      62.000000              0.000000      69.290000
50%     179.000000              0.000000      94.575000
75%     270.000000              0.000000     126.000000
max     543.000000            391.000000    5400.000000

       required_car_parking_spaces  total_of_special_requests
count                 119390.000000              119390.000000
mean                       0.062518                   0.571363
std                        0.245291                   0.792798
min                        0.000000                   0.000000
25%                        0.000000                   0.000000
50%                        0.000000                   0.000000
75%                        0.000000                   1.000000
max                        8.000000                   5.000000

[8 rows x 21 columns]
```

1. Data Cleaning

```
# missing values check
data.isnull().sum()
```

```
index                                   0
hotel                                   0
is_canceled                             0
lead_time                               0
arrival_date_year                       0
arrival_date_month                      0
arrival_date_week_number                0
arrival_date_day_of_month               0
stays_in_weekend_nights                 0
stays_in_week_nights                    0
adults                                  0
children                                4
babies                                  0
meal                                    0
country                               488
market_segment                          0
distribution_channel                    0
is_repeated_guest                       0
previous_cancellations                  0
previous_bookings_not_canceled          0
reserved_room_type                      0
assigned_room_type                      0
booking_changes                         0
deposit_type                            0
agent                               16340
company                            112593
days_in_waiting_list                    0
customer_type                           0
adr                                     0
required_car_parking_spaces             0
total_of_special_requests               0
reservation_status                      0
reservation_status_date                 0
dtype: int64
```

```
# missing values fill
data['children'] = data['children'].fillna(0)
data['country'] = data['country'].fillna('Unknown')
data['agent'] = data['agent'].fillna(0)
data['company'] = data['company'].fillna(0)

# duplicate values remove
data.drop_duplicates(inplace=True)
```

1. Feature Engineering

```
# total guests
data['total_guests'] = data['adults'] + data['children'] +
data['babies']
```

```
# total nights calculate
data['total_nights'] = data['stays_in_weekend_nights'] +
data['stays_in_week_nights']
```

1. Exploratory Data Analysis (EDA)

```
# total bookings
len(data)
```

```
119390
```

```
# cancellation count
data['is_canceled'].value_counts()
```

```
is_canceled
0    75166
1    44224
Name: count, dtype: int64
```

```
# hotel types
data['hotel'].value_counts()
```

```
hotel
City Hotel      79330
Resort Hotel    40060
Name: count, dtype: int64
```

```
# average lead time
data['lead_time'].mean()
```

```
np.float64(104.01141636652986)
```

```
# month wise bookings
data['arrival_date_month'].value_counts()
```

```
arrival_date_month
August       13877
July         12661
May          11791
October      11160
April        11089
June         10939
September    10508
March         9794
February      8068
November      6794
December      6780
January       5929
Name: count, dtype: int64
```

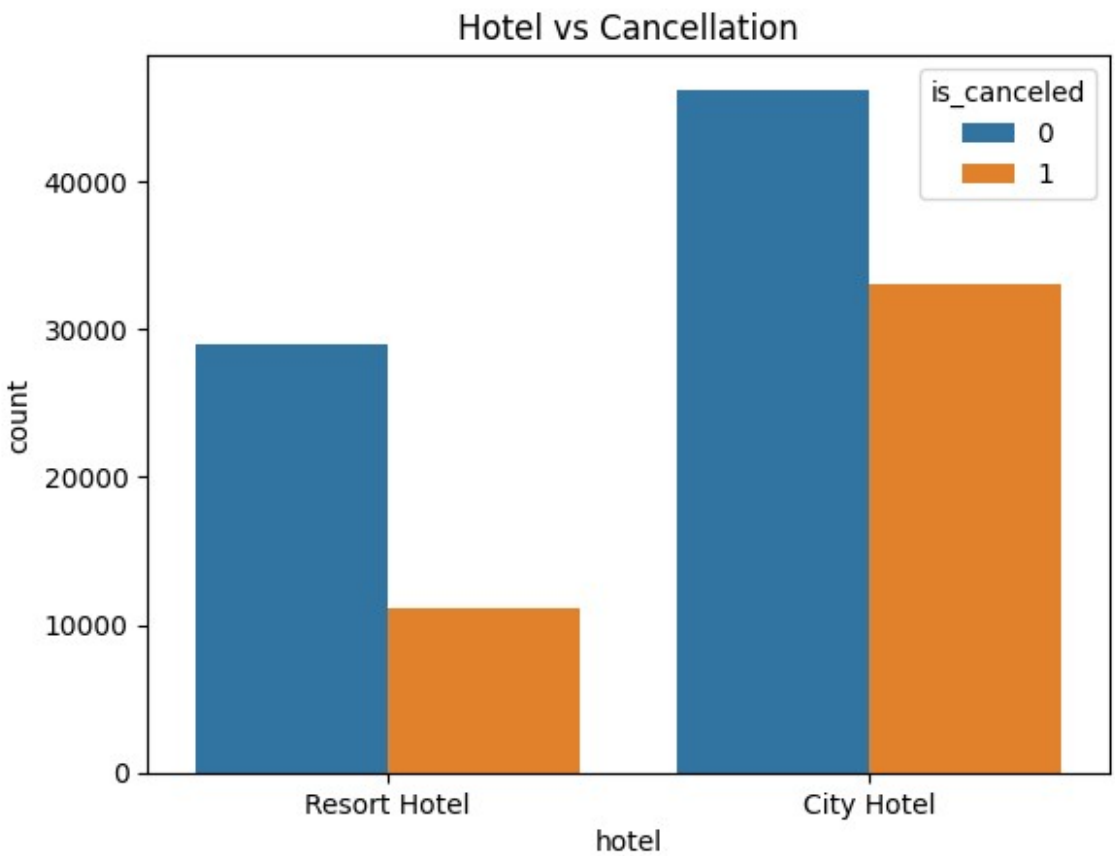
```
# top countries
data['country'].value_counts().head()
```

```
country
PRT    48590
GBR    12129
FRA    10415
ESP     8568
DEU     7287
Name: count, dtype: int64
```

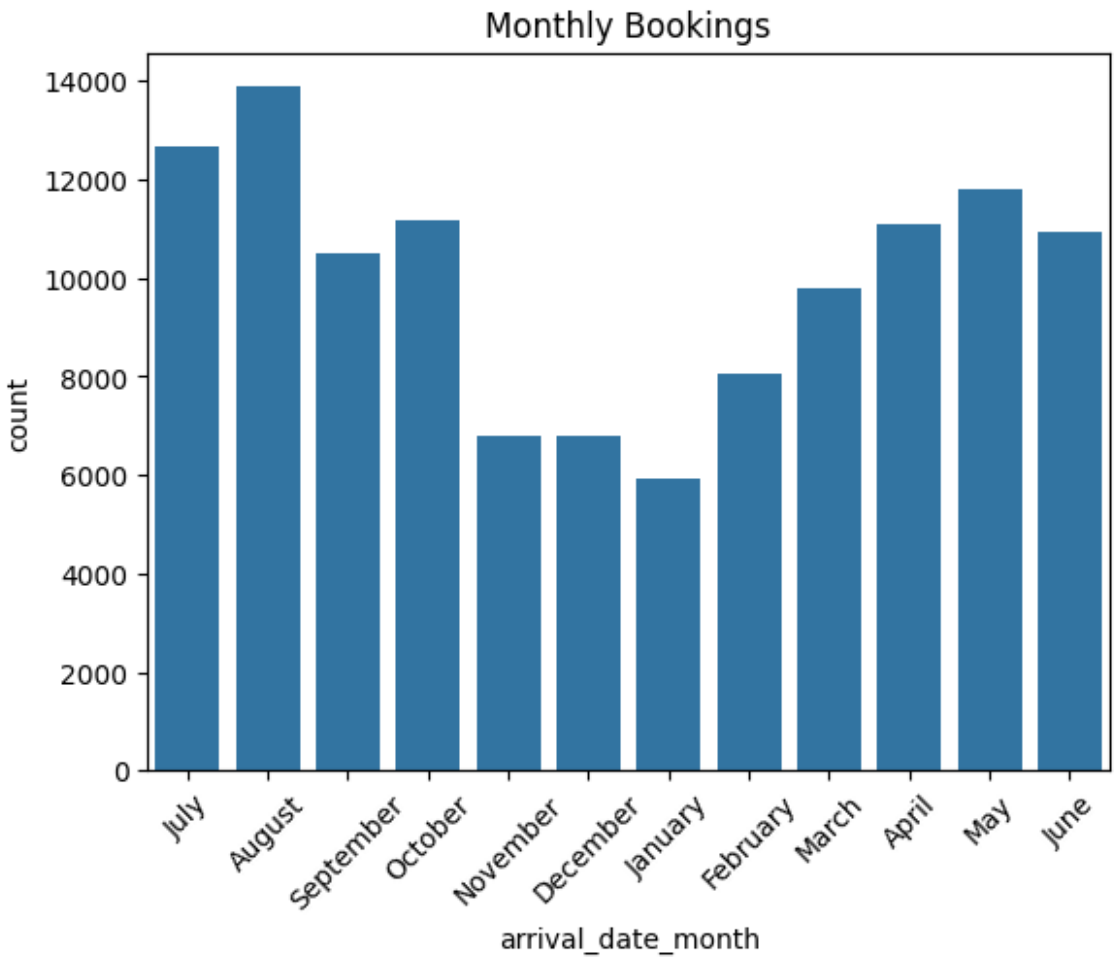
```
# cancellation rate
(data['is_canceled'].sum()/len(data))*100
```

```
np.float64(37.041628277075134)
```

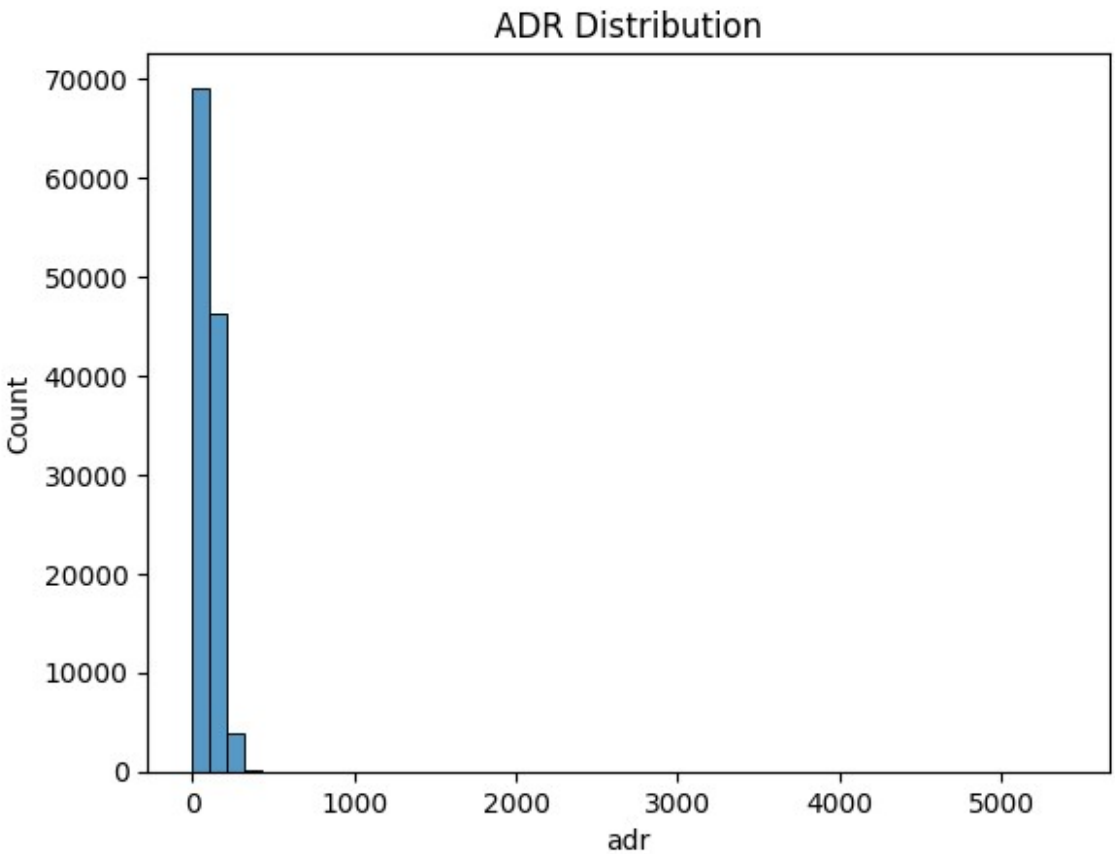
```
sns.countplot(x='hotel', hue='is_canceled', data=data)
plt.title("Hotel vs Cancellation")
plt.show()
```



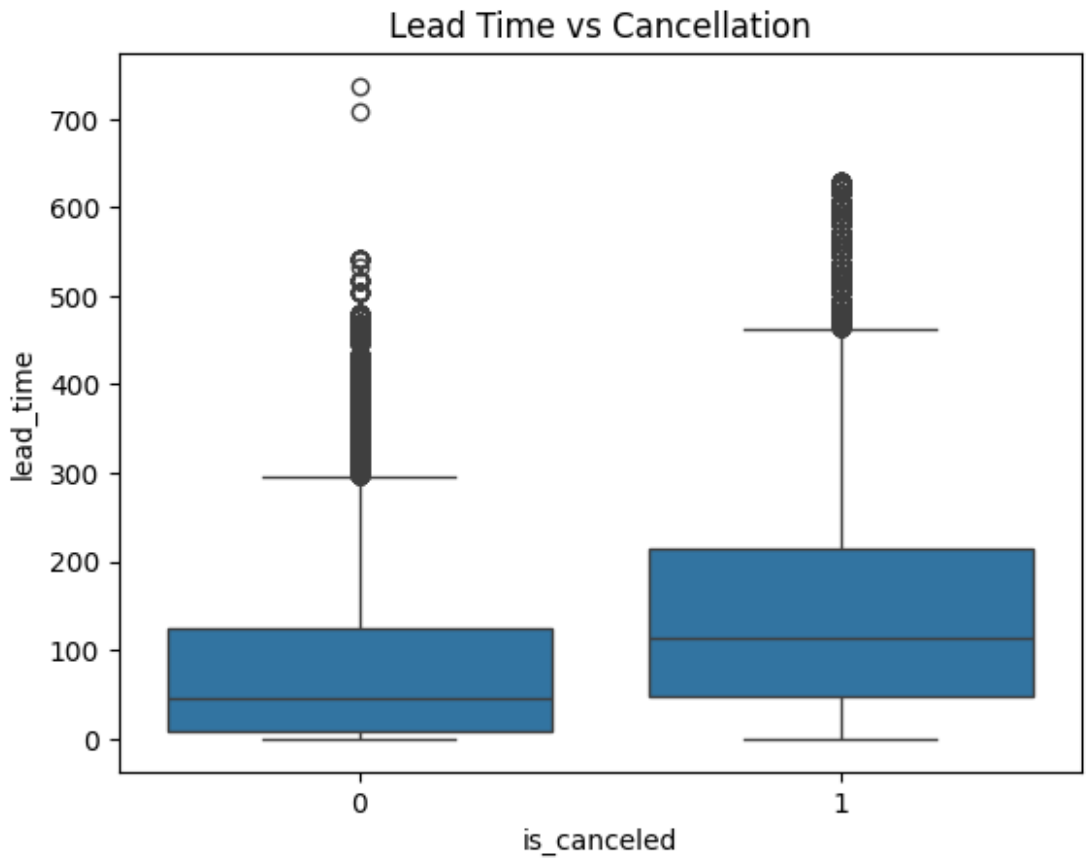

```
# month trend
sns.countplot(x='arrival_date_month', data=data)
plt.xticks(rotation=45)
plt.title("Monthly Bookings")
plt.show()
```




```
# adr distribution
sns.histplot(data['adr'], bins=50)
plt.title("ADR Distribution")
plt.show()
```

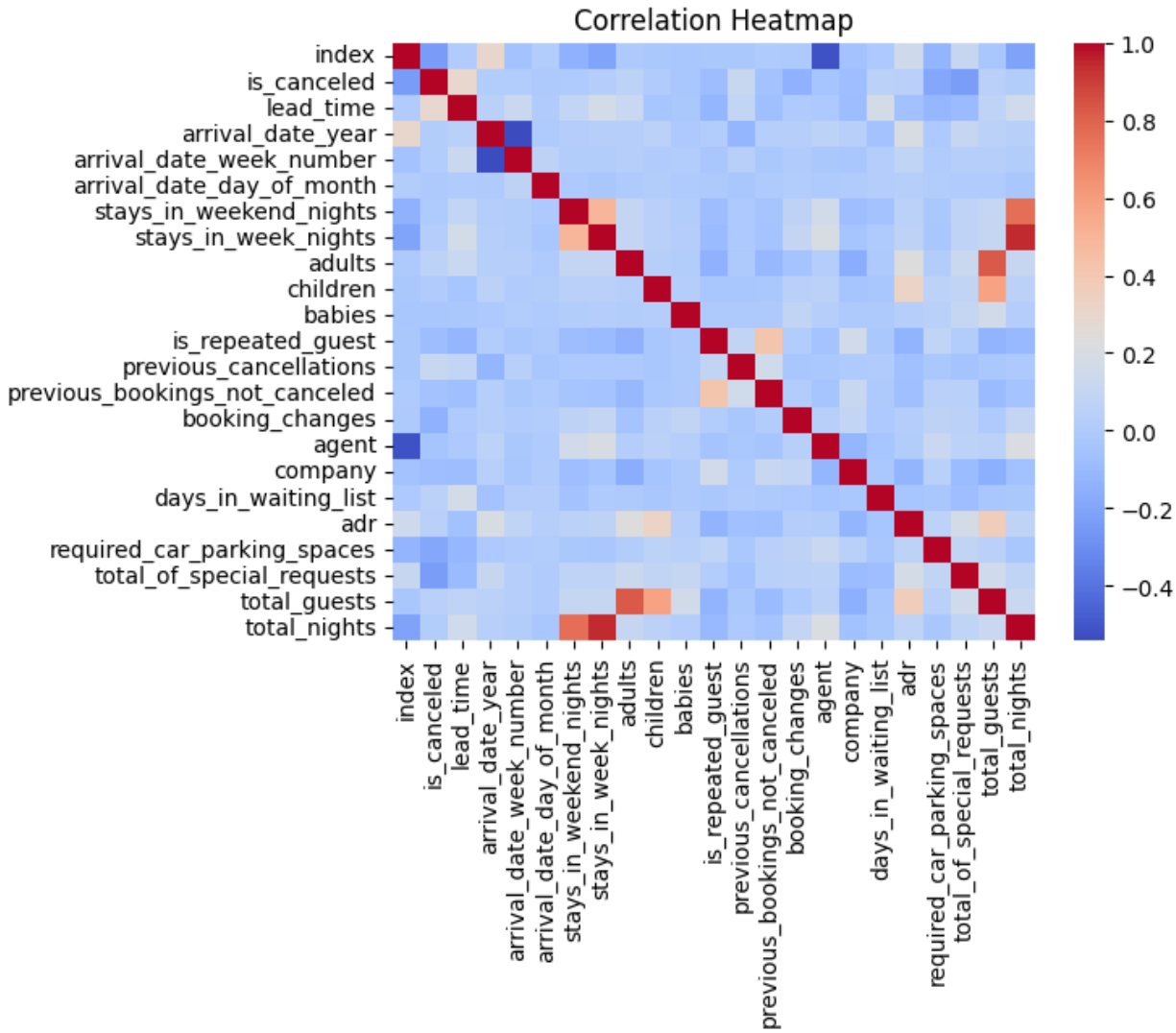



```
# lead time vs cancellation
sns.boxplot(x='is_canceled', y='lead_time', data=data)
plt.title("Lead Time vs Cancellation")
plt.show()
```



AVD analysis

```
# correlation heatmap
sns.heatmap(data.corr(numeric_only=True), cmap='coolwarm')
plt.title("Correlation Heatmap")
plt.show()
```




```
# room type vs adr
data.groupby('reserved_room_type')['adr'].mean()
```

```
reserved_room_type
A     90.796658
B     90.360447
C    160.216792
D    120.682882
E    124.540179
F    167.689651
G    175.996027
H    188.223111
L    124.666667
P      0.000000
Name: adr, dtype: float64
```

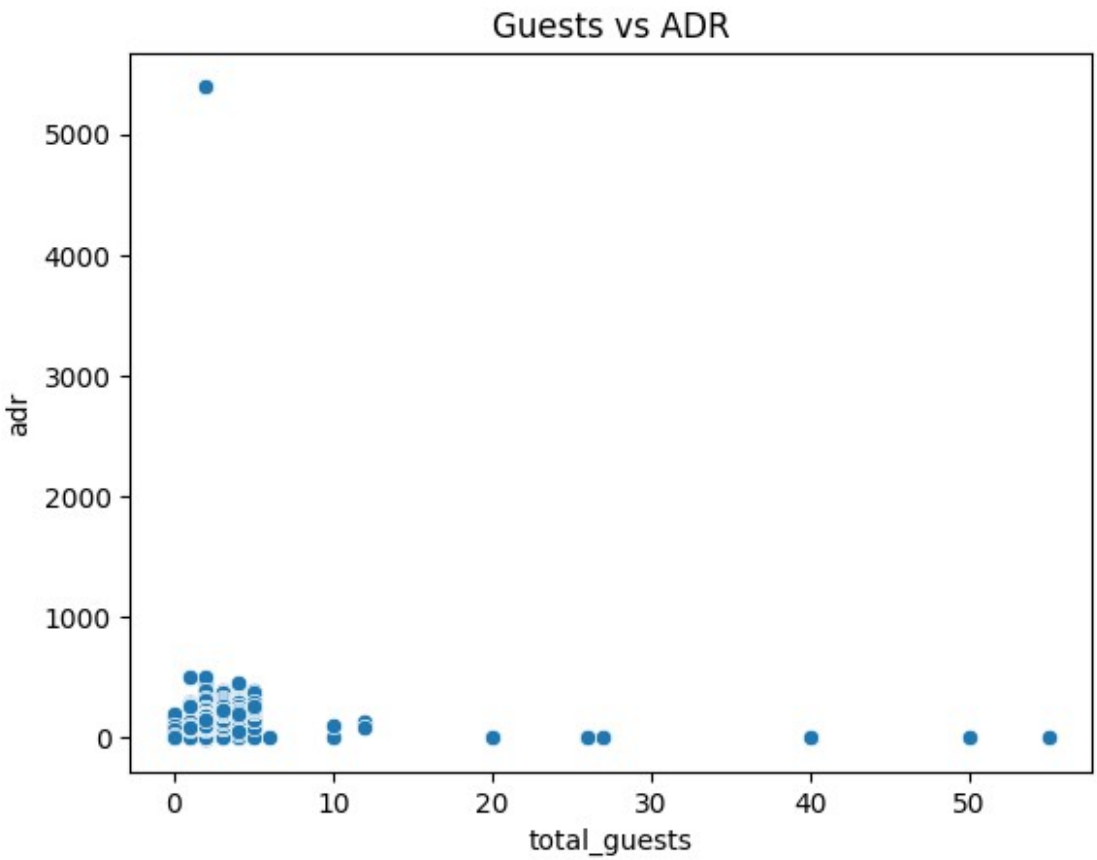
```python
# deposit type vs cancellation
data.groupby('deposit_type')['is_canceled'].mean()
```

```
deposit_type
No Deposit    0.283770
Non Refund    0.993624
Refundable    0.222222
Name: is_canceled, dtype: float64
```

```python
# special requests vs cancellation
data.groupby('total_of_special_requests')['is_canceled'].mean()
```

```
total_of_special_requests
0    0.477204
1    0.220249
2    0.220989
3    0.178614
4    0.105882
5    0.050000
Name: is_canceled, dtype: float64
```

```python
# total guests vs adr
sns.scatterplot(x='total_guests', y='adr', data=data)
plt.title("Guests vs ADR")
plt.show()
```


Guests vs ADR
adr
5000
4000
3000
2000
1000
0
0
10
20
30
40
50
total_guests